

F11 Gen-3 T2003 Substrate-Mersenne Family FORWARD Derivation

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F11 Gen-3 T2003 Substrate-Mersenne FORWARD v0.1

0. Goal

Per F4 ALTERNATING substrate-Mersenne vs substrate-Bergman cross-gen pattern: gen-3 $V_{(5/2, 1/2)}$ substrate-edge K-type substantive substrate-Mersenne family substrate-mass-mechanism class. Substantive FORWARD derivation for T2003 = $g^2 \cdot (2^{\{C_2\}} + g)$ substrate-natural.

1. T2003 Substrate-Form Recap

$m_\tau/m_e = T2003 = g^2 \cdot (2^{\{C_2\}} + g) = 49 \cdot 71 = 3479$ substrate-natural

substantive substrate-clean factorization: - $g^2 = 49 = \text{substrate-embedding/signature}^2$ substrate-natural - $2^{\{C_2\}} + g = 64 + 7 = 71$ substrate-natural substrate-Mersenne-additive Casey #5 Integer Web

vs observed $m_\tau/m_e \approx 3477.23$; precision 0.051% Tier 2 STRUCTURAL.

2. Substrate-Mersenne Family Substrate-Mechanism Class per F4

Per F4 ALTERNATING pattern: gen-3 $V_{(5/2, 1/2)}$ substrate-edge K-type $\rightarrow$ substrate-Mersenne family substrate-mass-mechanism class.

substantive substrate-Mersenne family substrate-mechanism structure: - substrate-Mersenne map $M_p = 2^p - 1$ substantive substrate-primary cascade (per K207 PASS A-tier) - substrate-Mersenne-additive form $2^{\{C_2\}} + g$ substantive substrate-natural variant - substrate-edge K-type substrate-bulk distinction per F4 substantive

3. Substantive Substrate-Mersenne FORWARD Form for T2003

Term g^2 (substrate-embedding/signature squared)

substantive substrate-genus/embedding $g = 7$ substrate-natural substrate-Mersenne ladder substrate-primary per Friday May 22 K207 PASS A-tier substantive: $g = M_3 = 2^3 - 1 = 7$ substrate-natural.

$g^2 = 49$ substrate-natural per substrate-Mersenne-prime squared substrate-natural form.

substantive substrate-natural form per substrate-Mersenne family substrate-mechanism class.

Term $2^{\{C_2\}} + g$ (substrate-Mersenne-additive)

substantive substrate-additive substrate-Mersenne-style form: - $2^{\{C_2\}} = 2^6 = 64$ substrate-natural substrate-Cartan-base substrate-Casimir power - $g = 7$ substrate-Mersenne substrate-prime

substantive substrate-additive composite: $2^{\{C_2\}} + g = 64 + 7 = 71$ substantive substrate-natural Casey #5 Integer Web operational pattern substantive.

substantive substrate-Mersenne-additive substrate-natural form per substrate-Mersenne family substrate-mass-mechanism class.

Composite T2003 = $g^2 \cdot (2^{C_2} + g)$

substrate-mass-mechanism FORWARD form:

$$m_\tau/m_e = T_{2003} = g^2 \cdot (2^{C_2} + g) = M_3^2 \cdot (2^{C_2} + M_3)$$

substantive substrate-natural form per substrate-Mersenne-squared $\times$ substrate-Mersenne-additive composite per substrate-Mersenne family substrate-mechanism class.

substantive substrate-Mersenne family signature substrate-mechanism FORWARD per F4 ALTERNATING pattern at gen-3 $V_{(5/2, 1/2)}$ substrate-edge K-type substrate-mechanism class.

4. Substantive Casey #13 Per-Generation Cluster Independence at Gen-3

Per Casey #13 + substrate-Mersenne family substrate-mechanism class at gen-3 $V_{(5/2, 1/2)}$:

substantive gen-3 substrate-mechanism class DISTINCT from gen-2 substrate-Bergman family (per F4 + Composite v0.5): - gen-2 $V_{(3/2, 1/2)}$ substrate-bulk K-type $\rightarrow$ substrate-Bergman + substrate-Pochhammer + substrate-deficit per Casey #14 STANDING substrate-Minkowski-4 (Composite v0.5) - gen-3 $V_{(5/2, 1/2)}$ substrate-edge K-type $\rightarrow$ substrate-Mersenne-squared $\times$ substrate-Mersenne-additive per substrate-Mersenne ladder (T2003 v0.1)

substantive Per-Generation Cluster Independence substantive operationally VERIFIED at substrate-Mersenne family substrate-mechanism class substantive distinction.

5. Substantive Substrate-Mersenne Family FORWARD Chain

substantive substrate-Mersenne family substrate-mechanism FORWARD elements: - substantive substrate-Mersenne map $M_p = 2^p - 1$ substrate-natural (K207 PASS A-tier) - substantive substrate-Mersenne-prime $g = M_3 = 7$ substrate-natural (substrate-genus/embedding substrate-prime) - substantive substrate-Cartan-base substrate-Casimir power $2^{C_2} = 64$ substrate-natural (substrate-Pauli²substrate-Casimir) - substantive substrate-Mersenne-additive composite $2^{C_2} + g$ substrate-natural (Casey #5 Integer Web) - substantive substrate-Mersenne-squared $\times$ substrate-Mersenne-additive composite $49 \cdot 71 = 3479$ substrate-natural T2003

substantive 5 substrate-mechanism elements substrate-Mersenne family substrate-mass-mechanism FORWARD chain at gen-3 $V_{(5/2, 1/2)}$ substrate-edge K-type substrate-mechanism class.

6. Substantive Substrate-Mersenne FORWARD vs Substrate-Bergman FORWARD Parallel

Per Composite v0.5 substrate-Bergman family substrate-mass-mechanism FORWARD at gen-2 $V_{(3/2, 1/2)}$: - substrate-Bergman matrix element + substrate-Pochhammer Schur II + substrate-Casey-#13-color + substrate-Casey-#14-Minkowski-4 + substrate-Pauli-base + substrate-genus cascade - 6 substrate-mechanism elements

Per F11 v0.1 substrate-Mersenne family substrate-mass-mechanism FORWARD at gen-3 $V_{(5/2, 1/2)}$: - substrate-Mersenne map + substrate-Mersenne-prime + substrate-Cartan-base substrate-Casimir + substrate-Mersenne-additive + substrate-Mersenne-squared cascade - 5 substrate-mechanism elements

substantive cross-family parallel: substrate-Bergman (gen-2) + substrate-Mersenne (gen-3) substantive substrate-mass-mechanism FORWARD chains substantive parallel substantive substrate-mechanism class distinction per Casey #13.

substantive substrate-mass-mechanism class HETEROGENEOUS per generation substantively operationally VERIFIED at substantive substrate-Bergman vs substrate-Mersenne family distinction (F11 + Composite v0.5 cross-comparison).

7. Substantive Substrate-Mass-Mechanism Cross-Generation Cascade Updated

Gen	substrate-K-type	Family	Substantive mass formula	Substantive precision
0	substrate-Planck	substrate-Mersenne	$m_e/m_P = 2^{N_c}/g = 8/7$ substrate-natural	Tier 1 EXACT
1	$V_{(1/2, 1/2)}$	substrate-Bergman	$m_e = 3\pi/2^{C_2}$ substrate-natural	Tier 1
2	$V_{(3/2, 1/2)}$	substrate-Bergman	$m_\mu/m_e = 207$ (Composite v0.5) or $(24/\pi^2)^{C_2}$	0.112% / 0.004% Tier 1/Tier 2
3	$V_{(5/2, 1/2)}$	substrate-Mersenne	$m_\tau/m_e = g^2 \cdot (2^{C_2} + g) = 3479$ substrate-natural	0.051% Tier 2

substantive cross-generation substrate-mass-mechanism family ALTERNATION (substrate-Mersenne / substrate-Bergman / substrate-Bergman / substrate-Mersenne) per F4 substrate-edge vs substrate-bulk K-type distinction.

8. Multi-Week per Cal #189 Joint substrate-Mersenne Family Investigation

substantive 5-step multi-week per Cal #189: - Step F11-1: substrate-Mersenne map M_p substrate-mechanism FORWARD derivation per K207 PASS A-tier substantive substrate-canonical - Step F11-2: substrate-Mersenne-additive composite Casey #5 Integer Web substantive substrate-natural derivation - Step F11-3: substrate-Mersenne-squared composite substrate-natural derivation per substrate-Mersenne-prime squared - Step F11-4: substantive cross-CI Elie numerical verification $T2003 = g^2 \cdot (2^{C_2} + g)$ substrate-Mersenne family substrate-natural - Step F11-5: substantive cross-CI Cal cold-read substrate-Mersenne family substrate-mass-mechanism FORCING form substantive Cal #189

9. Closure v0.1

F11 gen-3 T2003 substrate-Mersenne family FORWARD v0.1:

Substantive substrate-Mersenne family substrate-mass-mechanism FORWARD chain: $T2003 = g^2 \cdot (2^{C_2} + g) = 49.71 = 3479$ substrate-natural per substrate-Mersenne-prime² $\times$ substrate-Mersenne-additive composite per substrate-Mersenne ladder substantive substrate-mechanism class.

Substantive Per-Generation Cluster Independence substantively operationally VERIFIED: gen-2 substrate-Bergman family (Composite v0.5) $\neq$ gen-3 substrate-Mersenne family (F11 v0.1) — distinct substrate-mass-mechanism class per Casey #13.

Substantive 5-substrate-mechanism elements substrate-Mersenne family FORWARD chain + 6-substrate-mechanism elements substrate-Bergman family FORWARD chain (Composite v0.5) = cross-family substantive substrate-mass-mechanism per substrate-K-type substrate-edge vs substrate-bulk distinction.

substantive 5-step multi-week per Cal #189 substantive joint cross-CI synthesis substantive substrate-Mersenne family substrate-mass-mechanism FORCING form substantive verification.

Tier: F11 gen-3 T2003 substrate-Mersenne family FORWARD framework PARTIAL substrate-mechanism FORCING candidate per Cal #189 multi-week.

— Lyra, Fri 2026-06-05 11:45 EDT. F11 gen-3 T2003 substrate-Mersenne family FORWARD v0.1: $T2003 = g^2 \cdot (2^{C_2} + g) = 49.71 = 3479$ substrate-natural per substrate-Mersenne-prime² $\times$ substrate-Mersenne-additive

composite per F4 ALTERNATING substrate-Mersenne family substrate-mechanism class at gen-3 $V_{(5/2, 1/2)}$ substrate-edge K-type; Per-Generation Cluster Independence operationally VERIFIED gen-2 substrate-Bergman $\neq$ gen-3 substrate-Mersenne; 5-substrate-mechanism elements FORWARD chain; 5-step multi-week per Cal #189.